

3.6 QFT

Recap CFT & QM $\partial_\mu = \frac{\partial}{\partial x^\mu}$ $\partial^\mu = \frac{\partial}{\partial x_\mu}$

$$\phi = \phi(t, x, y, z) = \phi(t, \vec{x}) = \phi(x)$$

$$(\phi(t, \vec{x}) = \phi(t, x))$$

Lagrangian density $\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi)$

Action $S = \int L dt = \int \mathcal{L} d^4x$

Dimensions $[dx] = -1$

$$\left. \begin{array}{l} (E=m) \quad [d^4x] = -4 \\ [\mathcal{L}] = 4 \end{array} \right\} [S] = 0 \quad (?)$$

$$\delta S = \int d^4x \delta \mathcal{L}$$

$$= \int d^4x \frac{\delta \mathcal{L}}{\delta \phi} \delta \phi + \frac{\delta \mathcal{L}}{\delta (\partial_\mu \phi)} \delta (\partial_\mu \phi)$$

$$\dagger \quad \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0 \quad \text{E-L eqn.}$$

example. $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$ K-G $\phi = \text{a free KG field}$

$$= \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2 \quad (\text{scalar})$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi, \quad \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = \partial^\mu \phi = (\dot{\phi}, -\nabla \phi)$$

$$\dagger \quad 0 = \partial^\mu \partial_\mu \phi + m^2 \phi = (\partial^2 + m^2) \phi = \ddot{\phi} - \nabla^2 \phi + m^2 \phi$$

$$[\mathcal{L}] = 4 \quad [\partial] = 1 \quad [\phi] = 1 \quad [m] = 1$$

ex1. (vector) $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$

$\downarrow$ E-L

Maxwell's eqn.

$$\partial_i E^i = 0, \quad E_i = \epsilon_{ijk} \partial^j B^k$$

Lorentz invariance

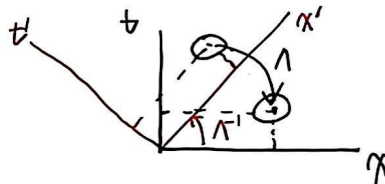
$$\mathcal{L} \xrightarrow{x^\mu \rightarrow x'^\mu} \mathcal{L} \quad (x'^\mu = \Lambda^\mu_\nu x^\nu)$$

$$\Lambda^\mu_\rho \eta^{\rho\sigma} \Lambda^\sigma_\tau = g^{\mu\nu} \quad \left(\Lambda^\mu_\nu \neq \Lambda^\mu_\nu \right)$$

x -axis rotation $\begin{pmatrix} 1 & & \\ & \cos \theta & -\sin \theta \\ & \sin \theta & \cos \theta \end{pmatrix} = \Lambda^\mu_\nu$

boost $\begin{pmatrix} \gamma & -\gamma v \\ -\gamma v & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix} = \Lambda^\mu_\nu$

$$\phi(x) \xrightarrow{\Lambda} \phi'(x) = \phi(\Lambda^{-1}x) \quad \Lambda^{-1}x \equiv (\Lambda^{-1})^\mu_\nu x^\nu$$



Noether's Theorem.

2025 QFT - Name - ID.



3.13 QFT

[Noether's Theorem]

Continuous Symmetry $\Rightarrow$ Conserved quantity.

eqn. of motion

$$\partial_\mu j^\mu = 0 \quad j \rightarrow \text{current}$$

$$Q = \int d^3x j^0 \Rightarrow \frac{dQ}{dt} = \dots$$

$$\text{if } \phi \rightarrow \phi + X \Rightarrow \delta L = 0$$

$$X = \delta\phi$$

$$0 = \delta L = \frac{\partial L}{\partial \phi} \delta\phi + \frac{\partial L}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi)$$

$$= E - L \text{ eqn.} + \partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \delta\phi \right)$$

$$\Rightarrow \partial_\mu j^\mu = \partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \delta\phi \right)$$

$$j^\mu = \frac{\partial L}{\partial(\partial_\mu \phi)} X$$

$$\text{if } \delta L = \partial_\mu F^\mu$$

$$\text{generally } j^\mu = \frac{\partial L}{\partial(\partial_\mu \phi)} X - F^\mu$$

$$\text{eq.1. } \alpha^\mu \rightarrow \alpha^\mu - \epsilon^\mu$$

$$\phi \rightarrow \phi + \epsilon^\mu \partial_\mu \phi \quad (\text{Taylor})$$

$$X = \epsilon^\mu \partial_\mu \phi$$

$$L \rightarrow L + \epsilon^\mu \partial_\mu L$$

$$\partial_\mu F^\mu = \delta L = \epsilon^\mu \partial_\mu L$$

$$= \partial_\mu \epsilon^\mu L$$

$$F^\mu = \epsilon^\mu L$$

$$j^\mu = \frac{\partial L}{\partial(\partial_\mu \phi)} X - F^\mu$$

$$= \epsilon^\nu \left[\frac{\partial L}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu L \right]$$

$$= \epsilon^\nu \underbrace{j^\mu_\nu}_{\text{energy-momentum tensor}}$$

$$j^\mu_\nu = \frac{\partial L}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta_\nu^\mu L \quad (\nu=0, \dots, 3)$$

$$\partial_\mu j^\mu_\nu = 0$$

$$E = \int d^3x T^{00}$$

$$P^i = \int d^3x T^{0i}$$

$$T^\mu_\nu$$

ex. K-Q Theory

eq.2 complex scalar field

$$\psi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$$

$$\psi^* = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2)$$

$$L = \partial_\mu \psi^* \partial^\mu \psi - V(\psi^* \psi)$$

$$V = m^2 \psi^* \psi + \frac{\lambda}{2} (\psi^* \psi)^2 + \dots$$

$$\dagger \quad \partial^2 \psi + \frac{\partial V}{\partial \psi} = 0 \quad (\partial^2 \equiv \partial_\mu \partial^\mu)$$

internal sym. $\psi \rightarrow e^{i\alpha} \psi$ * $\alpha \rightarrow \text{constant. (global)}$

$$\text{ex.2. } j^\mu = i \cancel{\partial^\mu} (\partial^\mu \psi^*) \psi - i \psi^* (\partial^\mu \psi)$$



3.13 QFT

[Hamiltonian]

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

$$\mathcal{H} = \pi \dot{\phi} - \mathcal{L} = \mathcal{H}(\phi, \pi)$$

$$H = \int d^3x \mathcal{H}$$

$$\text{Hamilton's eqn.} \begin{cases} \dot{\phi} = \frac{\partial \mathcal{H}}{\partial \pi} \\ \dot{\pi} = -\frac{\partial \mathcal{H}}{\partial \phi} \end{cases}$$

K-Q

$$\mathcal{L} = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - V$$

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}$$

$$\mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + V \quad (\text{Energy})$$

$$\begin{cases} [H, a^\dagger] = \omega a^\dagger \\ [H, a] = -\omega a \end{cases}$$

$$\text{--- } E + \omega$$

$$\text{--- } E$$

$$\text{--- } E - \omega$$

$\vdots$

$$\text{--- } |0\rangle$$

$$\nearrow a^\dagger \quad H(a^\dagger |E\rangle) = (E + \omega) a^\dagger |E\rangle$$

$$\searrow a$$

$$|n\rangle = (a^\dagger)^n |0\rangle.$$

$$a|0\rangle = 0$$

[Quantum Mechanics]

$$H = \frac{1}{2} p^2 + \frac{1}{2} \omega^2 q^2$$

$$p, q \rightarrow \hat{p}, \hat{q}$$

$$\rightarrow \hat{H} = \frac{1}{2} \hat{p}^2 + \frac{1}{2} \omega^2 \hat{q}^2$$

$$[\hat{q}, \hat{p}] = i$$

$$\text{SE} \quad i \frac{d}{dt} |\psi\rangle = H |\psi\rangle$$

$$H |E\rangle = E |E\rangle$$

$$|\psi\rangle = e^{-iEt} |E\rangle$$

$$H = \omega \left(\dots \right) \left(\dots \right)$$

$$= \omega a a^\dagger$$

(Ladder / creation op)

$$\begin{cases} q = \frac{1}{\sqrt{2}\omega} (a + a^\dagger) \\ p = i\sqrt{\frac{\omega}{2}} (a - a^\dagger) \end{cases}$$

$$[a, a^\dagger] = 1$$

$$\cancel{a^\dagger a}$$

$$H = \omega (a^\dagger a + \frac{1}{2}).$$

[QFT] — a real scalar field ϕ (KG) (free).

$$(\partial^2 + m^2) \phi = 0$$

$$\text{FT: } \phi(t, \mathbf{x}) = \int \frac{d^3p}{(2\pi)^3} e^{ip \cdot x} \tilde{\phi}(t, \mathbf{p})$$

$$\left[\frac{\partial^2}{\partial t^2} + (\mathbf{p}^2 + m^2) \right] \tilde{\phi} = 0$$

$$\mathbf{p}^2 + m^2 = \omega_{\mathbf{p}}^2$$

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (a_{\mathbf{p}} + a_{\mathbf{p}}^\dagger) e^{ip \cdot x} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (a_{\mathbf{p}} e^{ip \cdot x} + a_{\mathbf{p}}^\dagger e^{-ip \cdot x})$$

$$[\phi(x), \phi(y)] = 0 = [\pi(x), \pi(y)]$$

$$\pi(x) = \int \frac{d^3p}{(2\pi)^3} (i) \sqrt{\frac{\omega_{\mathbf{p}}}{2}} (a_{\mathbf{p}} - a_{\mathbf{p}}^\dagger) e^{ip \cdot x}$$

$$[\phi(x), \pi(y)] = i \delta^{(3)}(x - y)$$

$\Downarrow$

$$\text{ex. } [a_{\mathbf{p}}, a_{\mathbf{q}}] = 0 \quad [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] = 0, \quad [a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q}).$$



Hamiltonian. 补充 P2

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[a_p e^{ip \cdot x} + a_p^\dagger e^{-ip \cdot x} \right]$$

$$\pi(x) = \int \frac{d^3p}{(2\pi)^3} (-i) \left(\frac{\omega_p}{2} \right) \left[a_p e^{ip \cdot x} - a_p^\dagger e^{-ip \cdot x} \right] \quad \checkmark$$

$$H = \int d^3x \left[\frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 \right]$$

$$\int d^3x \pi^2 = - \int d^3x \frac{d^3p d^3q}{(2\pi)^6} (a_p e^{ip \cdot x} - a_p^\dagger e^{-ip \cdot x}) (a_q e^{iq \cdot x} - a_q^\dagger e^{-iq \cdot x}) \frac{\sqrt{\omega_p \omega_q}}{2}$$

$$= - \int d^3x \frac{d^3p d^3q}{(2\pi)^6} \left[a_p a_q e^{i(p+q) \cdot x} - a_p^\dagger a_q e^{i(q-p) \cdot x} - a_p a_q^\dagger e^{i(p-q) \cdot x} + a_p^\dagger a_q^\dagger e^{-i(p+q) \cdot x} \right] \frac{\sqrt{\omega_p \omega_q}}{2}$$

$$= - \int \frac{d^3p d^3q}{(2\pi)^6} \left[a_p a_q (2\pi)^3 \delta^{(3)}(p+q) - a_p^\dagger a_q (2\pi)^3 \delta^{(3)}(q-p) - a_p a_q^\dagger (2\pi)^3 \delta^{(3)}(p-q) + a_p^\dagger a_q^\dagger (2\pi)^3 \delta^{(3)}(p+q) \right] \frac{\sqrt{\omega_p \omega_q}}{2}$$

$$= \int \frac{d^3p}{(2\pi)^3} \left[(a_p^\dagger a_p + a_p a_p^\dagger) - (a_p a_{-p} + a_p^\dagger a_{-p}^\dagger) \right] \frac{\omega_p}{2}$$

$$\omega_p = \sqrt{p^2 + m^2}$$

$$\omega_{-p} = \sqrt{(-p)^2 + m^2} = \omega_p.$$

$$H = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_p (a_p^\dagger a_p + a_p a_p^\dagger) = \int \frac{d^3p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} [a_p, a_p^\dagger] \right)$$

$$\frac{1}{2} [a_p, a_p^\dagger] = \frac{1}{2} (a_p a_p^\dagger - a_p^\dagger a_p)$$

$$[a_p, a_q^\dagger] = (2\pi)^3 \delta^{(3)}(p-q).$$

$$= \int \frac{d^3p}{(2\pi)^3} \left(a_p^\dagger a_p + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right) \rightarrow \infty.$$

$$p=q \quad p-q=0$$

2∞ s.



3.20 QFT

$$H = \int d^3x \left[\frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 \right]$$

$$\int d^3x \pi^2 = - \int d^3x \frac{d^3p d^3q}{(2\pi)^6} \frac{\sqrt{\omega_p \omega_q}}{2} (a_{\vec{p}} e^{i\vec{p} \cdot \vec{x}} - a_{\vec{p}}^\dagger e^{-i\vec{p} \cdot \vec{x}}) (a_{\vec{q}} e^{i\vec{q} \cdot \vec{x}} - a_{\vec{q}}^\dagger e^{-i\vec{q} \cdot \vec{x}})$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{\omega_p}{2} [(a_{\vec{p}}^\dagger a_{\vec{p}} + a_{\vec{p}} a_{\vec{p}}^\dagger) - (a_{\vec{p}} a_{-\vec{p}} + a_{\vec{p}}^\dagger a_{-\vec{p}}^\dagger)]$$

$$H = \frac{1}{2} \int \frac{d^3p}{(2\pi)^3} \omega_p (a_{\vec{p}}^\dagger a_{\vec{p}} + a_{\vec{p}} a_{\vec{p}}^\dagger)$$

$$= \int \dots (a_{\vec{p}}^\dagger a_{\vec{p}} + \frac{1}{2} [a_{\vec{p}}, a_{\vec{p}}^\dagger])$$

$$= \int \dots (a_{\vec{p}}^\dagger a_{\vec{p}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0)) \leftarrow \infty$$

$$\downarrow$$

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_p a_{\vec{p}}^\dagger a_{\vec{p}}$$

$$:H: = \int \frac{d^3p}{(2\pi)^3} \omega_p a_{\vec{p}}^\dagger a_{\vec{p}} \quad \text{Normal Ordering}$$

$$[H, a_{\vec{p}}^\dagger] = \int \frac{d^3p}{(2\pi)^3} \omega_p [a_{\vec{p}}^\dagger a_{\vec{p}}, a_{\vec{p}}^\dagger] = \omega_p a_{\vec{p}}^\dagger$$

$$[H, a_{\vec{p}}] = -\omega_p a_{\vec{p}}$$

$$\omega_p = \sqrt{p^2 + m^2}$$

$$|p\rangle = a_{\vec{p}}^\dagger |0\rangle$$

$$= \sqrt{E_p} \quad (\text{SR})$$

$$H|p\rangle = \omega_p |p\rangle$$

$$= E_p$$

$|p\rangle$ 1 pti state (m, p)

$$|p_1, p_2, \dots, p_n\rangle = a_{\vec{p}_1}^\dagger \dots a_{\vec{p}_n}^\dagger |0\rangle \quad n\text{-ptl state}$$

$$|p, q\rangle = |q, p\rangle$$

$\hookrightarrow$ KG fields : bosons

Fock space: $|0\rangle, a^\dagger |0\rangle, a^\dagger a^\dagger |0\rangle, \dots$

(1) $\delta^{(3)}(0)$: Infrared divergence (IR)

$$(2\pi)^3 \delta^{(3)}(0) = \int d^3x$$

Universe in a box. (V) .

$$\rightarrow E_0 = V \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} \omega_p, \quad \varepsilon_0 = \frac{E_0}{V}$$

(2) $\int d^3p$ Ultraviolet divergence (UV)

$$\varepsilon_0 = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} \omega_p \quad \omega_p = \sqrt{p^2 + m^2}$$

cutoff scale Λ

Regularization

$$\text{const} = \frac{1}{2} (2\pi)^3 \delta^{(3)}(0)$$



1.2. 0.1.7

3.20 QF

Lorentz invariance?

$|0\rangle, \langle 0|0\rangle = 1$ vacuum

$|p\rangle, \langle p|q\rangle = (2\pi)^3 \delta^{(3)}(p-q)$ 3分量

$|p\rangle = A_p |p\rangle$

$\langle p|q\rangle = (2\pi)^3 A_p^* A_q \delta^{(3)}(p-q)$

$p^2 = p^\mu p_\mu, m^2, \int d^4p = LI \rightarrow \int d^4p \delta(p^2 - m^2) = LI$

$p_0 = \begin{cases} +\sqrt{p^2 + m^2} \\ -\sqrt{p^2 + m^2} \end{cases}$ $\int d^3p \delta(p_0^2 - p^2 - m^2) = LI$

$1 = \int d^3p \delta^{(3)}(p-q) = \int \frac{d^3p}{2E_p} 2E_p \delta^{(3)}(p-q) \rightarrow LI$

$|p\rangle = \sqrt{2E_p} |p\rangle, \langle p|q\rangle = (2\pi)^3 2E_p \delta^{(3)}(p-q)$

$a^\dagger(p) = \sqrt{2E_p} a_p^\dagger$

$\phi = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} [a(p)e^{ip\cdot x} + a^\dagger(p)e^{-ip\cdot x}]$

SP $\begin{cases} \phi, \pi & \text{operators: } x \\ | \rangle & \text{states: } t \end{cases}$ $i\frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$
 $|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$

HP $\begin{cases} \text{op: } t, x \end{cases}$ $\langle \psi(t) | O_S | \psi(t) \rangle$
 $= \langle \psi(0) | e^{iHt} O_S e^{-iHt} | \psi(0) \rangle$
 $= \langle \psi(0) | O_H(t) | \psi(0) \rangle$

$\frac{dO_H}{dt} = i[H, O_H]$ HE

1.2. 0.1.7

$\dot{\phi} = i[H, \phi(t, x)]$

$= \frac{i}{2} \int d^3y [\pi^2(t, y) + (\nabla_y \phi(t, y))^2 + m^2 \phi^2(t, y), \phi(t, x)]$

$\begin{cases} [\phi(t, x), \phi(t, y)] = 0 \\ [\phi(t, x), \pi(t, y)] = i\delta^{(3)}(x-y) \end{cases} \quad \begin{cases} \dot{\phi} = \pi \\ \dot{\pi} = \nabla^2 \phi - m^2 \phi = \ddot{\phi} \end{cases}$

$\ddot{\phi} - \nabla^2 \phi + m^2 \phi = 0$

$(\partial^2 + m^2)\phi = 0$ QKG-Field

$e^{iHt} a_p e^{-iHt} = e^{-iE_p t} a_p \quad \dots a_p^\dagger \dots = \dots a_p^\dagger$

$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} (a_p e^{-iE_p t + ip\cdot x} + a_p^\dagger e^{iE_p t - ip\cdot x})$

$= \int \dots (a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x}) \quad [\phi(t', x'), \phi(t, x)] = ?$

4.3 QFT

Quantum Field

3.2 QFT Causality

$$[\phi(t, \mathbf{x}), \phi(t', \mathbf{x}')] = 0 \quad \text{for spacelike separate} \quad (x-y)^2 < 0$$

$$x \& y \xrightarrow{LT} x = (t, \mathbf{x}), y = (t', \mathbf{x}')$$

1) $\xrightarrow{\quad} x = (0, \mathbf{x}), y = (t, \mathbf{x})$
 $\Delta(x-y) = [\phi(0, \mathbf{x}) - \phi(t, \mathbf{x})] \sim e^{-imt} - e^{imt} \quad (LI)$
 (m → mass) for timelike separate

2) field → ptl. → ?
 Propagators 传播子 → Feynmann ~
 $D(x-y) \equiv \langle 0 | \phi(x) \phi(y) | 0 \rangle \quad (HP)$
 $\underbrace{y, R}_{x, D}$

$$D(x-y) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_p} e^{-ip \cdot (x-y)}$$

$$\Delta(x-y) = \int \sim (e^{ip(y-x)} - e^{ip(x-y)})$$

$$D(x-y) - D(y-x) = \Delta(x-y)$$

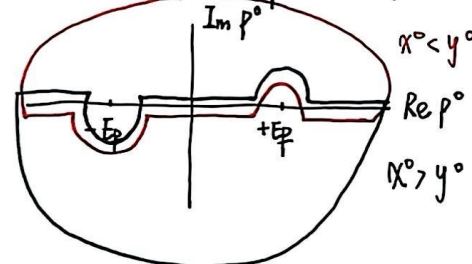
Feynmann Propagator

$$\Delta_F(x-y) = \langle 0 | T \phi(x) \phi(y) | 0 \rangle \quad T \text{ time-ordering op.}$$

(happen first → right side)

CLAIM $= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} e^{-ip(x-y)} \quad (\text{with the prescription})$

$$p^2 - m^2 = (p^0)^2 - E_p^2 \quad \text{pole at } p^0 = \pm E_p$$



$$p^0 = \pm E_p \xrightarrow{\text{Residue}} \pm \frac{1}{2E_p}$$

case 1: $x^0 > y^0, p^0 \rightarrow -i\infty, e^{-ip^0(x^0-y^0)} \rightarrow 0$

$$\int \frac{d^3 p}{(2\pi)^3} \frac{-2\pi i}{2E_p} i e^{-iE_p(x^0-y^0) + ip(\mathbf{x}-\mathbf{y})} = D(x-y)$$

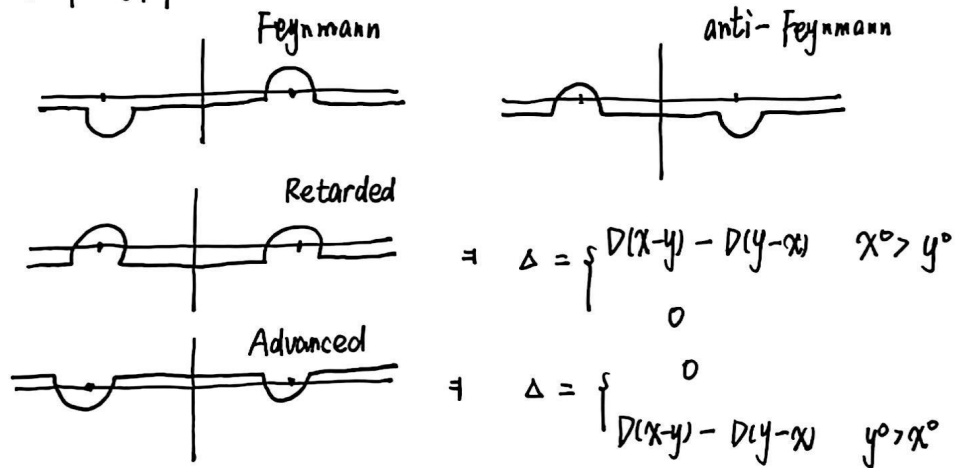
case 2: $x^0 < y^0, p^0 \rightarrow i\infty$

$$\int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_p} e^{-ip(x-y)} = D(y-x)$$

$$\Delta_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip(x-y)} \quad i\epsilon - \text{prescription}$$



3.27 QFT



Feynmann Propagator: Green's func. of K-G op.

$$\text{ex)} \quad (\partial^2 + m^2) \Delta_F(x-y) = -i \delta^{(4)}(x-y)$$

It's easy to prove.

$$\frac{1}{p^2 - m^2} = \frac{1}{(p^0)^2 - E_p^2} = \dots (x).$$

$$\begin{aligned} (\partial^2 + m^2) \Delta_F(x-y) &= (\partial_\mu^2 - \nabla^2 + m^2) \Delta_F(x-y) \\ &= \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} (-p^2 + m^2) e^{-ip \cdot (x-y)} \\ &= -i \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x-y)} \\ &= -i \delta^{(4)}(x-y). \end{aligned}$$

Interacting real scalar field (quantize)

$$\text{(free)} \quad \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$$

Equ of motion $\Rightarrow$ linear in field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \sum_{n=3}^{\infty} \frac{\lambda_n}{n!} \phi^n$$

interaction terms
if they are "small", then ~~perturbation~~ theory.

$$\begin{aligned} [\lambda_n] &= 4-n \\ [\lambda_n E^{n-4}] &= 0 \\ \underbrace{\begin{cases} \ll 1 \\ \gg 1 \end{cases}}_{\lambda_n \neq 4} & \end{aligned}$$

$\downarrow$ "Marginal"
 $\lambda_4: \lambda_4 \ll 1$
 $\lambda_{n \neq 4}$: dimensionful

$\hookrightarrow$ Perturbative QFT
 $\updownarrow$
Non-Per QCD

- $n=3$ $\lambda_3/E \xrightarrow{E \uparrow} \downarrow$
 $\hookrightarrow$ small at high energy "Relevant"
large at ...
- $n>4$ $\lambda_n E^{n-4} \xrightarrow{E \uparrow} \uparrow$ "Irrelevant"



4.3 QFT

Interacting Real Scalar Quantum Field

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \sum_{n=3}^{\infty} \frac{\lambda_n}{n!} \phi^n$$

① coupling constant λ_n (考虑全排列).

1) $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - \frac{\lambda}{4!} \phi^4$ ← self-interaction
 $\therefore \phi^4$ theory ($\lambda q \ll 1$)

($\mathbb{Z}_2$ symmetry: $\phi \rightarrow -\phi \rightarrow$ no ϕ^3 .) 3-interaction $\rightarrow$ Yukawa term

2) $\mathcal{L} = \partial_\mu \psi^* \partial^\mu \psi + \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - M^2 \psi^* \psi - \frac{1}{2} m^2 \phi^2 - g \psi^* \psi \phi$
 ψ : complex scalar $\phi \nabla \psi$
 ϕ : real scalar
 coupling constant: $g, [g]=1$: **Scalar Yukawa Theory**
 ($* g/m \ll 1, g/m \ll 1$)

Interaction Picture.

$H = H_0 + H_{int} \rightarrow$ interaction
 $\downarrow$
 H_P : i) send H to HP
 ii) leave H_{int} to SP $\rightarrow$ interaction picture

QM

SP: $|\psi(t)\rangle_S, O_S$

$i \frac{d}{dt} |\psi(t)\rangle_S = H |\psi(t)\rangle_S \Rightarrow |\psi(t)\rangle_S = e^{-iH(t-t_0)} |\psi(t_0)\rangle_S$
 $= U_S |\psi(t_0)\rangle_S$

HP: $|\psi\rangle_H, O_H(t) = e^{iH_0 t} O_S e^{-iH_0 t}$

$|\psi\rangle_H = e^{iH_0 t} |\psi(t)\rangle_S$

$\frac{d}{dt} O_H(t) = i[H_0, O_H]$

IP: $O_I(t) = e^{iH_0 t} O_S e^{-iH_0 t}$
 $|\psi(t)\rangle_I = e^{iH_0 t} |\psi(t)\rangle_S$

$i \frac{d}{dt} |\psi(t)\rangle_S = H_I |\psi(t)\rangle_S$

$\hookrightarrow i \frac{d}{dt} (e^{-iH_0 t} |\psi(t)\rangle_I) = (H_0 + H_{int})_S e^{-iH_0 t} |\psi(t)\rangle_I$

$\Rightarrow \cancel{H_0 e^{-iH_0 t} |\psi(t)\rangle_I} + i e^{-iH_0 t} \frac{d}{dt} |\psi\rangle_I = \cancel{H_0 e^{-iH_0 t} |\psi\rangle_I} + H_{int} e^{-iH_0 t} |\psi\rangle_I$

$\Rightarrow i \frac{d}{dt} |\psi\rangle_I = e^{iH_0 t} H_{int} e^{-iH_0 t} |\psi\rangle_I$

$H_I \rightarrow i \frac{d}{dt} |\psi\rangle_I = H_I |\psi\rangle_I ; \frac{dO_I}{dt} = i[H_0, O_I]$

$$i \frac{d}{dt} |\psi\rangle_2 = H_2 |\psi\rangle_2$$

$$|\psi(t)\rangle_2 = U(t, t_0) |\psi(t_0)\rangle_2$$

$$i) U(t_0, t_0) = 1$$

$$ii) U(t_2, t_1) U(t_1, t_0) = U(t_2, t_0)$$

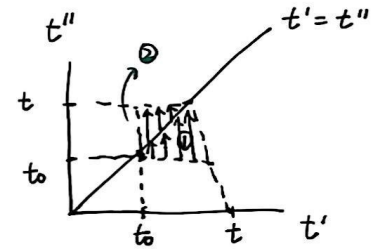
$$[H_1(t), H_2(t)] \neq 0$$

$$U(t, t_0) = T \exp \left\{ -i \int_{t_0}^t H_2(t') dt' \right\} \quad \text{Dyson's Formula}$$

↑ time-ordering op.

$$\begin{aligned} i \frac{d}{dt} [T \exp \left\{ -i \int_{t_0}^t H_2(t') dt' \right\}] &= i T \left[(-i) H_2(t) \exp \left\{ -i \int_{t_0}^t H_2(t') dt' \right\} \right] \\ &= T [H_2(t) \exp \{ \dots \}] = a^\dagger a a^\dagger \\ &= H_2(t) T \exp \{ \dots \} = H_2(t) U = a^\dagger a a^\dagger \end{aligned}$$

$$T(O_1(t_1) O_2(t_2)) = \begin{cases} O_1(t_1) O_2(t_2) & t_1 > t_2 \\ O_2(t_2) O_1(t_1) & t_2 > t_1 \end{cases}$$



Perturbative Interaction (H_2 is small)

$$\begin{aligned} U(t, t_0) &= T \left[1 + (-i \int_{t_0}^t H_2(t') dt') + \frac{1}{2!} (-i \int_{t_0}^t H_2(t') dt')^2 + \dots \right] \\ &= T \left[1 - i \int_{t_0}^t H_2(t') dt' + \frac{(-i)^2}{2!} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t') H_2(t'') + \dots \right] \\ &= 1 - i \int_{t_0}^t H_2(t') dt' - \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t') H_2(t'') + \dots \end{aligned}$$

$$\begin{aligned} \leftarrow T \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t') H_2(t'') &= \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t') H_2(t'') + \int_{t_0}^t dt' \int_{t'}^t dt'' H_2(t'') H_2(t') \\ &\downarrow \\ 2 \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t') H_2(t'') & \\ \boxed{\int_{t_0}^t dt' \int_{t'}^t dt'' H_2(t'') H_2(t')} & \text{ (Diagram 1: } t'' > t' \text{)} \\ = \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t'') H_2(t') & \text{ (Diagram 2: } t'' < t' \text{)} \\ = \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' H_2(t') H_2(t'') & \text{ (Diagram 3: rename } t', t'' \text{)} \end{aligned}$$



4.10 QFT Lecture 6.

Example: Scalar Yukawa theory

$$\mathcal{L} = \partial_\mu \psi^\dagger \partial^\mu \psi + \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - M^2 \psi^\dagger \psi - \frac{1}{2} m^2 \phi^2 - g \psi^\dagger \psi \phi \quad ([g]=1, g \ll M, m, E)$$

Free $\sim$: $\mathcal{L}_\psi = |\partial \psi|^2 - M^2 |\psi|^2$

$$(\tilde{a}_{\vec{p}} = \frac{\partial \psi}{(2\pi)^3}) \left\{ \begin{aligned} \psi &= \int \tilde{d^3p} (b_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + c_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}) \frac{1}{\sqrt{2E_{\vec{p}}}} & b^\dagger \rightarrow 'b'', c^\dagger \rightarrow 'c'' \\ \psi^\dagger &= \int \tilde{d^3p} (c_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + b_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}) \frac{1}{\sqrt{2E_{\vec{p}}}} & 'b' : \text{anti-ptl of 'c'} \\ \pi &= i \int \tilde{d^3p} \sqrt{E_{\vec{p}}} (b_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}} - c_{\vec{p}} e^{i\vec{p}\cdot\vec{x}}) = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = \dot{\psi}^* \\ \pi^\dagger &= \dots \end{aligned} \right.$$

$$[b_{\vec{p}}, b_{\vec{q}}^\dagger] = [c_{\vec{p}}, c_{\vec{q}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$$

S-Y theory: $H_{\text{int}} = g \int d^3x \psi^\dagger \psi \phi$

$$\left\{ \begin{aligned} \phi &\sim a + a^\dagger : \text{"}\phi \text{ ptls"} \rightarrow \text{meson} \\ \psi &\sim b + c^\dagger : \text{create "c", destroy "b"} \\ \psi^\dagger &\sim c + b^\dagger : \text{" " "b", " " "c"} \end{aligned} \right.$$

nucleon anti-nucleon

$H_{\text{int}} \supset b^\dagger c^\dagger a$ $b \leftarrow \text{meson} \rightarrow c$ Decay of Meson

$H_{\text{int}}^2 \supset b^\dagger c^\dagger a c b a^\dagger$ $c \rightarrow b, b \rightarrow c$ Scattering $N-\bar{N}$

Decay of Meson

Initial State $|i\rangle \Rightarrow$ free theory $\left\{ \begin{aligned} t \rightarrow -\infty \\ t \rightarrow +\infty \end{aligned} \right.$
Final State $|f\rangle$
 $\lim_{t_i \rightarrow -\infty} \langle f | U(t_f, t_i) | i \rangle \equiv \langle f | S | i \rangle : \text{"S-matrix"}$

$$|i\rangle = \sqrt{2E_p} a_p^\dagger |0\rangle$$

$$|f\rangle = \sqrt{4E_{q_1} E_{q_2}} b_{q_1}^\dagger c_{q_2}^\dagger |0\rangle$$

$$\langle f | S | i \rangle = \langle f | [1 - i \int_{-\infty}^{+\infty} dt H_{\text{int}}(t) + \dots] | i \rangle$$

$$= \langle f | 1 | i \rangle - i \langle f | \int_{-\infty}^{+\infty} dt H_{\text{int}}(t) | i \rangle + \dots$$

$$= -i \langle f | \int_{-\infty}^{+\infty} dt H_{\text{int}}(t) | i \rangle$$

$$\Rightarrow \langle f | S | i \rangle = -i g \langle f | \int d^4x \psi^\dagger(x) \psi(x) \phi(x) | i \rangle$$

$$\langle f | \psi^\dagger \psi \phi | i \rangle \sim \langle 0 | c b (b^\dagger + c) (\psi + c^\dagger) (a + a^\dagger) a^\dagger | 0 \rangle$$

$$\sim \langle 0 | c b b^\dagger c^\dagger a a^\dagger | 0 \rangle$$

"be lazy"



$$\langle f | S | i \rangle = -ig \langle 0 | \sqrt{4E_{q_1} E_{q_2}} c_{q_2} b_{q_1} \int d^4x \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_k}} \int \frac{d^3k_1}{(2\pi)^3} \frac{1}{\sqrt{2E_{k_1}}} \int \frac{d^3k_2}{(2\pi)^3} \frac{1}{\sqrt{2E_{k_2}}} b_{k_2}^\dagger e^{ik_2 \cdot x} c_{k_1}^\dagger e^{ik_1 \cdot x} a_k e^{-ik \cdot x} \sqrt{2E_p} a_p^\dagger | 0 \rangle$$

$$= -ig \langle 0 | \int d^4x \int \frac{d^3k d^3k_1 d^3k_2}{(2\pi)^9} \frac{\sqrt{E_{q_1} E_{q_2}}}{E_k E_{k_1} E_{k_2}} e^{i(k_1+k_2) \cdot x} e^{-ik \cdot x} c_{q_2} b_{q_1} b_{k_2}^\dagger c_{k_1}^\dagger a_k a_p^\dagger | 0 \rangle$$

$$\rightarrow [a_k, a_p^\dagger] | 0 \rangle = (2\pi)^3 \delta^{(3)}(p-k) | 0 \rangle$$

$$= -ig \langle 0 | \int d^4x \int \frac{d^3k_1 d^3k_2}{(2\pi)^6} \frac{\sqrt{E_{q_1} E_{q_2}}}{E_{k_1} E_{k_2}} e^{i(k_1+k_2) \cdot x} e^{-ip \cdot x} c_{q_2} b_{q_1} b_{k_2}^\dagger c_{k_1}^\dagger | 0 \rangle$$

$$= -ig \langle 0 | \int d^4x \int \frac{d^3k_1 d^3k_2}{(2\pi)^6} \frac{\sqrt{E_{q_1} E_{q_2}}}{E_{k_1} E_{k_2}} e^{i(k_1+k_2) \cdot x} e^{-ip \cdot x} c_{q_2} c_{k_1}^\dagger b_{q_1} b_{k_2}^\dagger | 0 \rangle$$

$$\rightarrow [b_{q_1}, b_{k_2}^\dagger] | 0 \rangle = (2\pi)^3 \delta^{(3)}(q_1 - k_2) | 0 \rangle$$

$$= -ig \langle 0 | \int d^4x \int \frac{d^3k_1}{(2\pi)^3} \frac{\sqrt{E_{q_2}}}{E_{k_1}} e^{i(k_1+q_1) \cdot x} e^{-ip \cdot x} c_{q_2} c_{k_1}^\dagger | 0 \rangle$$

$$\rightarrow [c_{q_2}, c_{k_1}^\dagger] | 0 \rangle = (2\pi)^3 \delta^{(3)}(q_2 - k_1) | 0 \rangle$$

$$= -ig \langle 0 | \int d^4x e^{i(q_2+q_1-p) \cdot x} | 0 \rangle$$

$$= -ig (2\pi)^4 \delta^{(4)}(q_2+q_1-p) \quad \text{4-momentum conservation}$$

$$= -ig (2\pi)^4 \delta^{(4)}(q_2+q_1-p) \quad \text{4-Momentum Conservation}$$

$$p = q_1 + q_2 \quad \text{and} \quad \langle f | S | i \rangle \neq 0$$

Wick's Theorem : $T \rightarrow : \dots :$



4.17 QFT Lecture 7.

Dyson's formula: T

what we want: $;$ $;$ $\rightarrow$ Wick's theorem

Contraction

$$\overline{\phi\psi} \equiv T(\phi\psi) - :\phi\psi:$$

$$\phi = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx}) \equiv \phi^+ + \phi^-$$

$$\overline{\phi(x)\phi(y)}$$

$$x^0 > y^0: T(\phi(x)\phi(y)) = \phi(x)\phi(y)$$

$$= (\phi_x^+ + \phi_x^-)(\phi_y^+ + \phi_y^-)$$

$$= \phi_x^+ \phi_y^+ + \phi_x^+ \phi_y^- + \phi_x^- \phi_y^+ + \phi_x^- \phi_y^-$$

$$\underbrace{[\phi_x^+, \phi_y^-]}_{\text{contraction}} + \phi_y^- \phi_x^+$$

$$= \phi_x^+ \phi_y^+ + \phi_y^- \phi_x^+ + \phi_x^- \phi_y^+ + \phi_x^- \phi_y^- + [\phi_x^+, \phi_y^-]$$

$$:\phi(x)\phi(y): = \phi_x^+ \phi_y^+ + \phi_y^- \phi_x^+ + \phi_x^- \phi_y^+ + \phi_x^- \phi_y^-$$

$$\Rightarrow \overline{\phi(x)\phi(y)} = [\phi_x^+, \phi_y^-]$$

$$x^0 < y^0: \overline{\phi(x)\phi(y)} = [\phi_y^+, \phi_x^-]$$

$$\Rightarrow \overline{\phi(x)\phi(y)} = \begin{cases} [\phi^+(x), \phi^-(y)] & x^0 > y^0 \\ [\phi^+(y), \phi^-(x)] & x^0 < y^0 \end{cases} \begin{matrix} = D(x-y) \\ = D(y-x) \end{matrix} \equiv \Delta_F(x-y)$$

propagator $D(x-y) \equiv \langle 0 | \phi(x)\phi(y) | 0 \rangle = \langle 0 | [\phi^+(x), \phi^-(y)] | 0 \rangle = [\phi^+(x), \phi^-(y)]$

$$\hookrightarrow \overline{\phi(x)\phi(y)} = \Delta_F(x-y)$$

EX. $\overline{\psi(x)\psi^\dagger(y)} = \Delta_F(x-y)$, $\overline{\psi^\dagger(x)\psi(y)} = \Delta_F(y-x)$
others = 0

Wick's thm.

$$\phi_i \equiv \phi(x_i), T(\phi_1 \phi_2 \dots \phi_n) = :\phi_1 \phi_2 \dots \phi_n: + \text{all possible contractions}$$

$$\text{eg. } T(\phi_1 \phi_2 \phi_3 \phi_4) = :\phi_1 \phi_2 \phi_3 \phi_4: + 6 \overline{\phi_1 \phi_2} : \phi_3 \phi_4: + 3 \overline{\phi_1 \phi_3} : \phi_2 \phi_4: + 3 \overline{\phi_1 \phi_4} : \phi_2 \phi_3:$$

Suppose: Wick's thm is true for $\phi_2 \phi_3 \dots \phi_n$

add $\phi_1 (x_1^0 > x_{2,3,\dots,n}^0)$

$$T(\phi_1 \phi_2 \dots \phi_n) = (\phi_1^+ + \phi_1^-) (: \phi_2 \phi_3 \dots \phi_n: + : \text{all } \phi_i \phi_j:)$$

Every time we pass $\phi_{m>2,3,\dots,n}$, we'll get $\overline{\phi_1 \phi_m}$

Nucleon - Nucleon Scattering

$$|i\rangle = \sqrt{2E_{p_1}} \sqrt{2E_{p_2}} b_{p_1}^\dagger b_{p_2}^\dagger |0\rangle \equiv |p_1, p_2\rangle$$

$$|f\rangle = \sqrt{2E_{p_1}} \sqrt{2E_{p_2}} b_{p_1}^\dagger b_{p_2}^\dagger |0\rangle \equiv |p_1, p_2\rangle \quad \leftarrow \mathcal{O}(g)$$

$$\langle f | S | i \rangle = \langle f | 1 + i \int_{-\infty}^{+\infty} H_2 dt | i \rangle - \frac{i}{2} \langle f | T \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} H_2(t_1) H_2(t_2) dt_1 dt_2 | i \rangle + \dots$$

$\mathcal{O}(g)$ term: 0 $\leftarrow$ no scattering

Leading-order contribution: $\mathcal{O}(g^2)$ term

$$-\frac{g^2}{2} \int d^4x_1 \int d^4x_2 \langle f | T \psi_1^\dagger \psi_1 \phi_1 \psi_2^\dagger \psi_2 \phi_2 | i \rangle$$

$2^6 = 64$

only relevant term $\rightarrow \overline{\phi_1 \phi_2} : \psi_1^\dagger \psi_1 \psi_2^\dagger \psi_2 :$



4.17 Lec 7.

$$\langle f | : \psi_1^\dagger \psi_1 \psi_2^\dagger \psi_2 : | i \rangle = \int \frac{d^3 k_1 d^3 k_2 d^3 k_3 d^3 k_4}{(2\pi)^{12}} \frac{\sqrt{16 E_{k_1} E_{k_2} E_{k_3} E_{k_4}}}{\sqrt{16 E_{k_1} E_{k_2} E_{k_3} E_{k_4}}} \langle 0 | b_{p'_2}^\dagger b_{p'_1}^\dagger b_{k_1}^\dagger b_{k_2}^\dagger b_{k_3} b_{k_4} b_{p_1}^\dagger b_{p_2}^\dagger | 0 \rangle e^{i(k_1 \cdot x_1 + k_2 \cdot x_2 - k_3 \cdot x_1 - k_4 \cdot x_2)}$$

$$\begin{aligned} \langle 0 | \dots | 0 \rangle &= [b_{k_4}, b_{p_1}^\dagger] \langle 0 | b_{p'_2}^\dagger b_{p'_1}^\dagger b_{k_1}^\dagger b_{k_2}^\dagger b_{k_3} b_{p_2}^\dagger | 0 \rangle + \langle 0 | b_{p'_2}^\dagger b_{p'_1}^\dagger b_{k_1}^\dagger b_{k_2}^\dagger b_{k_3} b_{p_2}^\dagger | 0 \rangle [b_{k_4}, b_{p_1}^\dagger] \\ &= [b_{k_4}, b_{p_1}^\dagger] [b_{k_3}, b_{p_2}^\dagger] \langle 0 | b_{p'_2}^\dagger b_{p'_1}^\dagger b_{k_1}^\dagger b_{k_2}^\dagger | 0 \rangle + [b_{k_4}, b_{p_2}^\dagger] [b_{k_3}, b_{p_1}^\dagger] \langle 0 | b_{p'_2}^\dagger b_{p'_1}^\dagger b_{k_1}^\dagger b_{k_2}^\dagger | 0 \rangle \\ &= [b_{k_4}, b_{p_1}^\dagger] [b_{k_3}, b_{p_2}^\dagger] \left([b_{p'_1}, b_{k_1}^\dagger] \langle 0 | b_{p'_2}^\dagger b_{k_2}^\dagger | 0 \rangle + [b_{p'_1}, b_{k_2}^\dagger] [b_{p'_2}, b_{k_1}^\dagger] \right) + \dots \end{aligned}$$

$$\begin{aligned} &= [b_{k_4}, b_{p_1}^\dagger] [b_{k_3}, b_{p_2}^\dagger] [b_{p'_1}, b_{k_1}^\dagger] [b_{p'_2}, b_{k_2}^\dagger] + [b_{k_4}, b_{p_1}^\dagger] [b_{k_3}, b_{p_2}^\dagger] [b_{p'_1}, b_{k_2}^\dagger] [b_{p'_2}, b_{k_1}^\dagger] \\ &+ [b_{k_4}, b_{p_2}^\dagger] [b_{k_3}, b_{p_1}^\dagger] [b_{p'_1}, b_{k_1}^\dagger] [b_{p'_2}, b_{k_2}^\dagger] + [b_{k_4}, b_{p_2}^\dagger] [b_{k_3}, b_{p_1}^\dagger] [b_{p'_1}, b_{k_2}^\dagger] [b_{p'_2}, b_{k_1}^\dagger] \\ &= (2\pi)^{12} \left[\delta^{(3)}(k_4 - p_1) \delta^{(3)}(k_3 - p_2) \delta^{(3)}(p'_1 - k_1) \delta^{(3)}(p'_2 - k_2) + (k_1 \leftrightarrow k_2) + (p_1 \leftrightarrow p_2) + (k_1 \leftrightarrow k_2, p_1 \leftrightarrow p_2) \right] \end{aligned}$$

$$\langle f | : \psi_1^\dagger \psi_1 \psi_2^\dagger \psi_2 : | i \rangle = e^{i(p'_1 \cdot x_1 + p'_2 \cdot x_2 - p_1 \cdot x_1 - p_2 \cdot x_2)} + (p'_1 \leftrightarrow p'_2) + (p_1 \leftrightarrow p_2) + (p'_1 \leftrightarrow p'_2, p_1 \leftrightarrow p_2)$$

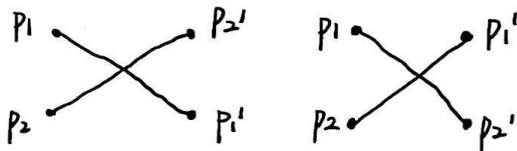
$$\mathcal{O}(g^2) = -\frac{g^2}{2} \int d^4 x_1 d^4 x_2 \{ \dots \} \int \frac{d^4 k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{ik \cdot (x_1 - x_2)}$$

$$= -g^2 \int d^4 x_1 d^4 x_2 \left\{ e^{i(p'_1 - p_2) \cdot x_1} e^{i(p'_2 - p_1) \cdot x_2} + e^{i(p'_1 - p_1) \cdot x_1} e^{i(p'_2 - p_2) \cdot x_2} \right\} \int \frac{d^4 k}{(2\pi)^4} \frac{i e^{ik \cdot (x_1 - x_2)}}{k^2 - m^2 + i\epsilon}$$

$$= -g^2 \int \frac{d^4 k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \cdot (2\pi)^4 \cdot (2\pi)^4 \left[\delta^{(4)}(p'_1 - p_2 + k) \delta^{(4)}(p'_2 - p_1 - k) + \delta^{(4)}(p'_1 - p_1 + k) \delta^{(4)}(p'_2 - p_2 - k) \right]$$

$$= -g^2 (2\pi)^4 \int \frac{i}{(p_2 - p'_1)^2 - m^2 + i\epsilon} \delta^{(4)}(p'_2 - p_1 - p'_1 + p_2) + \dots$$

$$= -g^2 (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p'_1 - p'_2) \left[\frac{i}{(p_1 - p'_1)^2 - m^2 + i\epsilon} + \frac{i}{(p_1 - p'_2)^2 - m^2 + i\epsilon} \right]$$



$$\mathcal{O}(g^2) \text{ for } \begin{matrix} p_1 p_2 & p'_1 p'_2 \\ NN \rightarrow & NN \end{matrix}$$

- strength $\sim g^2$
- E-M conservation
- ΔF for 2 configurations

→ Feynman Diagrams.

